

# Matthew Fong

Portfolio: <https://matthewfong3.github.io/>

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## Projects:

### Deserted Game (Unreal Engine 5)

- Implemented Enhanced Player Input Subsystem for player controls, such as movement, crouching, jumping, shooting, reloading, aim down sight (ADS) and smooth transition logic between player states
- Developed weapon struct system allowing implementation of various different types of weapons
- Implemented different weapon gun fire modes (automatic fire, semi-automatic fire, burst fire) on different weapons, allowing the player to switch a weapon's fire mode
- Scripted a time trial minigame feature in the shooting range level
- Implemented a weapon buy system similar to Call of Duty Zombies mode, where the player can spend points to purchase weapons around the map or replenish ammo

### Interlinked (Unreal Engine 5)

- Designed the code architecture of the player's 6 ability spells
- Programmed 2 of the 6 player's ability spells with UE5's Gameplay Ability System (GAS)
- Designed and implemented 2 of 3 enemy AI's behavior trees
- Implemented procedural walk animations with UE5's control rig and locomotor
- Implemented animation blueprint and animation montages for Torture Nexus, an AI enemy

### TPS Multiplayer Demo (Unreal Engine 5)

- Programmed player controls with the Enhanced Input Subsystem, such as movement, sprinting, jumping, crouching, shooting, aiming down sight (ADS), and reloading with C++
- Programmed replication of line tracing when shooting, applying damage, reloading, sound effects and particle effects from server across all clients through C++
- Programmed the logic behind updating weapon ammunition when reloading
- Programmed the animation blueprint for the third person character
- Designed and whiteboxed a simple level to demo the multiplayer prototype

### Configura (Godot)

- Programmed the extraction of the base model meshes into a single scene and accessory meshes into their own respective scenes
- Programmed the feature to dynamically loading new mesh assets and unloading previously selected assets in the character creator to improve performance and optimization
- Implemented dynamic camera movement to focus on the section of the character model that is currently being edited
- Refactored existing codebase to improve readability and maintainability
- Created and updated code documentation for the internal team and exterior documentation for developers and artists

## Skills:

**Programming Languages:** C++, C#, GDScript, JavaScript

**Software:** Unreal Engine 5, Blender, Godot, Unity, Visual Studio, VS Code, Autodesk Maya

**Concepts:** Object-Oriented Programming, Agile Development, Version Control, Data Structures & Algorithms, UI/UX Design, RESTful-API

**Tools/Frameworks:** Github, Trello, Jira, HTML5, CSS3, ES6, NodeJS, ReactJS, Websockets (socket.io), ExpressJS, MVC framework, Sass, Flexbox

## Education:

### Rochester Institute of Technology

Master of Science in Game Design and Development

Rochester, NY

Expected May 2027

**Relevant Coursework:** Advanced Topics in Game Dev (Unreal Engine 5), AI for Gameplay, Production Studio, Gameplay and Prototyping, Game Development Processes, Game Design

**Overall GPA:** 4.00/4.00

Bachelor of Science in Game Design and Development, Immersion in Music

May 2018

**Relevant Coursework:** Game Design & Development, Game Graphics Programming, Data Structures & Algorithms, Game Dev & Algorithm Problem Solving, Level Design, 3D Animation & Asset Production

**Honors and Awards:** Summa Cum Laude, Dean's List Fall 2014 - Spring 2018

## Work Experience:

### Magic Spell Studios

Traverse Creative Technologist Founders Program Developer

Rochester, NY

May-August 2026

- Developing Configura, an open-source character creation tool built in Godot
- Maintaining and expanding Configura's core functionality, including its character creator add-on
- Developing new features
- Improving existing systems
- Collaborating with 3D artists to integrate character assets
- Promoting the project through online communities

### The Langham Fifth Avenue, New York

Guest Service Ambassador

New York, NY

May 2022-April 2024

- Collaborated with supervisors and team members in brainstorming new ideas to improve workflow efficiency
- Clear and prompt communication between different departments within the company, regarding special guest needs, requests, inquiries
- Listened to guest complaints, diffusing tense situations and implementing solutions to ensure maximum guest satisfaction
- Regularly following up with guests to gain feedback and ensure their stay and service provided was nothing less of perfection